A

No.	Time	Source	Destination	Protocol	Lengtl	Info
89	1.833842200	192.168.1.3	192.168.1.1	DNS	71	Standard query 0x9a9f HTTPS th.bing.com
90	1.834157700	192.168.1.3	192.168.1.1	DNS	71	Standard query 0x8404 A th.bing.com
92	1.841023500	192.168.1.1	192.168.1.3	DNS	340	Standard query response 0x8404 A th.bin

b

```
Time to Live: 128
Protocol: UDP (17)
Header Checksum: 0x0000 [validation disabled]
```

C

173	2.929096700	192.168.1.3	192.168.1.1	DNS	82	Standard query
174	2.930741700	192.168.1.1	192.168.1.3	DNS	114	Standard query

```
Time to Live: 64
Protocol: UDP (17)
Header Checksum: 0xb734 [validation disabled
```

D

```
User Datagram Protocol, Src Port: 53812, Dst Port: 53
    Source Port: 53812
    Destination Port: 53
```

E

```
User Datagram Protocol, Src Port: 53, Dst Port: 53812
    Source Port: 53
    Destination Port: 53812
```

F

```
Source Address: 192.168.1.3
Destination Address: 192.168.1.1
```

G

```
Domain Name System (query)
    Transaction ID: 0xb669
  > Flags: 0x0100 Standard query

Domain Name System (response)
    Transaction ID: 0xb669
  > Flags: 0x8180 Standard query response, No error
```

Transaction ID: Query = 0xb669, Response = 0xb669. The Transaction ID is a 16-bit identifier used to match a DNS response with its corresponding DNS query.

H

```
Domain Name System (response)
    Transaction ID: 0xb669
  > Flags: 0x8180 Standard query response, No error
```

The DNS Flags field has a length of **16 bits** (2 bytes).

I

```
Domain Name System (query)
    Transaction ID: 0xb669
    Flags: 0x0100 Standard query
        0... .... .... .... = Response: Message is a query

Domain Name System (response)
    Transaction ID: 0xb669
    Flags: 0x8180 Standard query response, No error
        1... .... .... .... = Response: Message is a response
```

QR value	Meaning
0	Query
1	Response

J

```
Flags: 0x8180 Standard query response, No error
    1... .... .... .... = Response: Message is a response
    .000 0... .... .... = Opcode: Standard query (0)
    .... .0.. .... .... = Authoritative: Server is not an authority for domain
    .... ..0. .... .... = Truncated: Message is not truncated
    .... ...1 .... .... = Recursion desired: Do query recursively
    .... .... 1... .... = Recursion available: Server can do recursive queries
    .... .... .0.. .... = Z: reserved (0)
    .... .... ..0. .... = Answer authenticated: Answer/authority portion was not authenticated by the server
    .... .... ...0 .... = Non-authenticated data: Unacceptable
    .... .... .... 0000 = Reply code: No error (0)
```

K

```
> Flags: 0x0100 Standard query
  Questions: 1
  Answer RRs: 0
  Authority RRs: 0
  Additional RRs: 0
```

L

```
> Flags: 0x8180 Standard query response, No error
  Questions: 1
  Answer RRs: 2
  Authority RRs: 0
  Additional RRs: 0
```